

1. A link to YouTube video explaining your contribution to the project (10 points).

<https://youtu.be/Ka0sSeJKYyM>

2. Discussion of your contribution using a slideset (10 points)

1. **Stage 1 – Ideation** Our team began by brainstorming a wide range of project ideas, eventually focusing on ways to make social games more accessible for visually impaired individuals, inspired by adaptive chess concepts. I proposed building a device that would allow visually impaired people to play UNO alongside sighted players, while another teammate suggested a poker-based concept. After consulting with Professor Ayca Tas and weighing the feasibility of each idea within our five-week timeline, we collectively decided to move forward with the UNO project.
2. **Stage 2 – Design & Sensor Selection** In the second stage, I took the lead on designing the system and identifying the appropriate sensors for the project. I tested each sensor individually to understand how it functioned and to verify compatibility with the Raspberry Pi before integrating anything further.
3. **Stage 3 – Hardware Integration & Debugging** The following stage involved connecting all selected components on a breadboard, assigning GPIO ports, and refining the code so that every sensor could operate together reliably. Despite the team having agreed on set meeting days and being informed of the open lab schedule, I was the only member who consistently attended in person which was required since the project could only be worked on physically in the lab. This time, though, proved valuable: I identified numerous bugs in the code and encountered hardware issues that required me to swap out the Raspberry Pi three times before achieving stable operation.
4. **Stage 4 – Task Delegation & Knowledge Transfer** Since my teammates were unable to contribute to the hands-on development, I redirected their efforts toward written deliverables such as the report and presentation slides, allowing me to stay focused on testing and refining the device. I also made it a priority to walk my teammates through the project explaining how the system worked and the logic behind the code so that they would be able to confidently present and answer questions from guests or non-engineering audiences.
5. **Stage 5 – Feedback & Iteration** During the fourth week, I received feedback regarding the speech output speed through the earphone. After extensive testing with a larger set of cards and several code adjustments, I traced the issue to the AI-based card detection occasionally failing to accurately identify a card's number or color, which caused the text-to-speech output to run faster than intended. Identifying this root cause allowed me to address it more precisely.
6. **Stage 6 – Final Deliverables & Quality Control** In the final week, I coordinated the completion of all remaining deliverables, including the report, presentation, and demo videos taking personal responsibility for recording, editing, and ensuring

everything met the required standard. I set clear deadlines for each teammate to help keep the team on track and submitted everything on time.

7. **Reflection on Leadership** Throughout the project, my role as team leader meant being accountable for the quality and correctness of every submission — reviewing files before deadlines, resubmitting any incorrect materials, delegating tasks appropriately, and stepping in when needed. While this came with added pressure, it significantly strengthened my time management, communication, and leadership skills, and reinforced the importance of understanding project requirements and delivering a product that truly meets them.

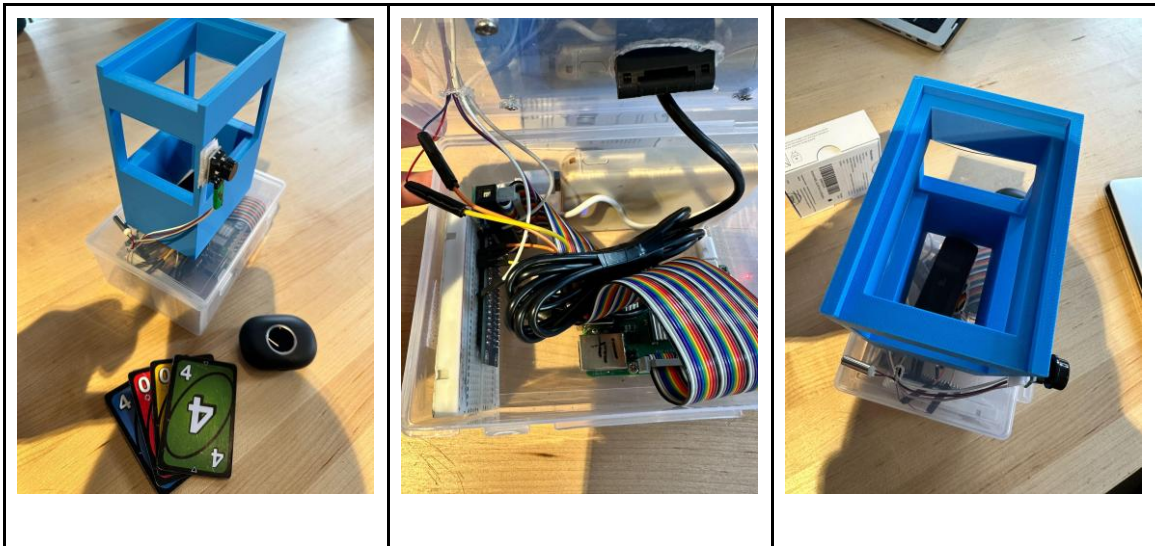


Table 1: Contribution: Project Design, Wiring, 3D Print, Assemble

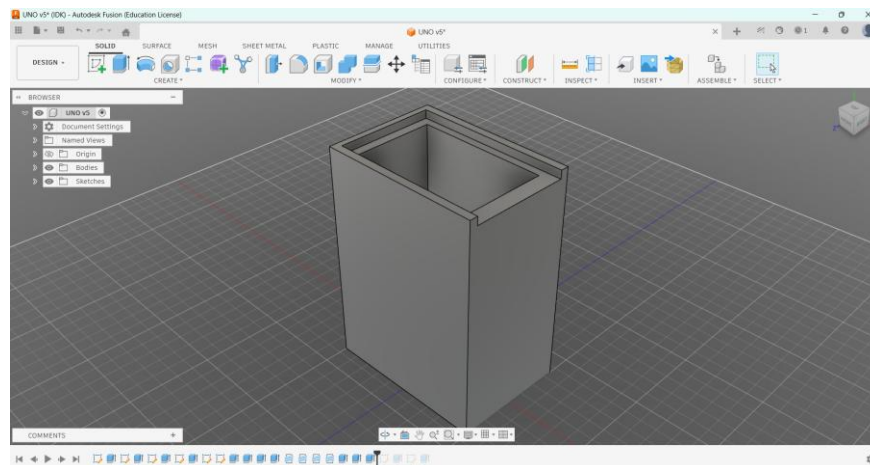


Table 2: First 3D Design

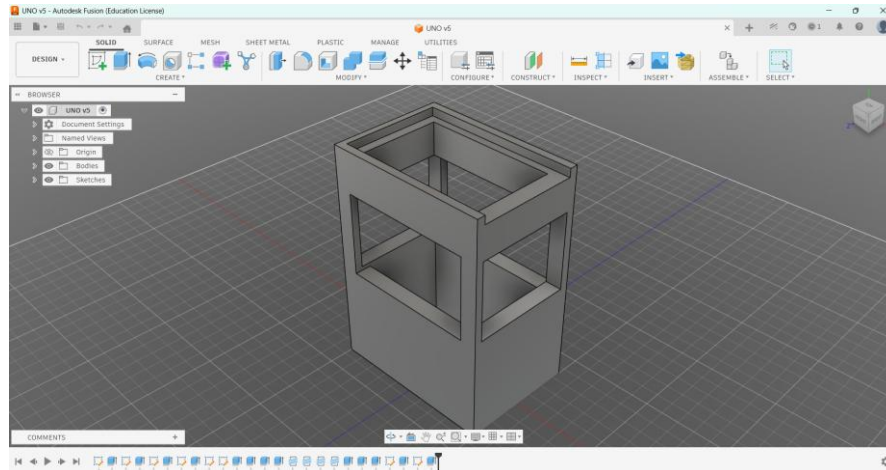


Table 3: Second 3D Design

Code (Zip file attached)

3. (Reflection on project experience, 4 points)

- For this project, most of the thing works as expected, however, the accuracy of the camera AI scan can be improve as well as the speed of the speech output from the earphones can be adjust slower. In the future, I learned that I should give my teammates a specific time frame for them to finish on time and not delaying the work over and over again.
- Most of the time is me going to the lab and work on the project since we can only work with the project while in class and I believe I did good in making sure the group delivered all submission on time and keep up with the tasks listed in the grant chart.
- The biggest tradeoffs maybe I might upset some of my teammates since I usually push them to meet with my expectation.
- The steps that worked well for me is coming to open lab and asked for the help of Sathwin and Lillian since some of the improvements, ideas are from them.
- I would dig deeper to make this project to be perfect if I were to redo this. For example, adding a database which can help the user to store the card without having to manually memorize how many cards the user has or may be add the features for right handed users as well, etc.

4. (Reflection on Design Process, 2 points)

- I believe every engineer should come up with a blueprint, lists out what steps to approach the problem, come up with an idea first so that he/she can save time on working on the project. Additionally, engineers should know what the users/customers requirements and ask the right and all the questions needed so that he/she can deliver the product precisely.

5. (Reflection on value for Customers, 2 points)

- We did talk to the customer to make sure we understand the problem clearly and precisely to deliver a correct product (Instructor: Dr. Ayca Tuzmen)
- We have a lot of feedback while in the process of developing a product such as make sure the product should be good looking, practical and user friendly as well as not to over-complicated the functionality of the product.
- We changed the position of the button so that the user can conveniently press the button while they put the card to be scanned
- If customer feedback is not collected before releasing a product, the team risks delivering something that does not meet user expectations, leading to poor user experience, low adoption, and damage to the team's reputation. Without real-world input, hidden bugs and usability issues only surface after launch, making them far more costly and time-consuming to fix. In the worst case, the product may need to be significantly redesigned, wasting the time and resources already invested

6. (Reflection on team work and skills, 2 points)

- Working in a team taught me that communication and accountability are just as important as technical skills. Throughout this project, I learned that not everyone will contribute equally, and as a leader, I had to adapt by redistributing tasks to keep the project moving forward. It also showed me the importance of making sure every team member understands the project, not just the person building it, so that everyone can confidently represent the work.
- Through this project, I discovered that my personal strengths lie in problem-solving, time management, and taking initiative — I was consistently present in the lab, debugging hardware issues, and making sure the project stayed on track even when I was working alone. However, I also recognized a weakness in my ability to delegate effectively early on.

Instead of pushing my teammates to be more involved from the start, I took on most of the technical work myself, which put a lot of pressure on me.

- On the technical side, I brought skills in hardware integration, sensor testing, coding, and debugging, which were essential in getting the project to work properly — especially when dealing with recurring Raspberry Pi issues and refining the AI-based card detection system. On the non-technical side, I contributed strong leadership, time management, and communication skills, which helped keep the team organized and on track throughout the five weeks.

- After reflecting on this project, I realize that I need to develop stronger delegation skills and learn how to hold my teammates more accountable from the start, rather than taking on extra work myself when things fall behind. On the technical side, I want to improve my understanding of AI and computer vision, since the card detection accuracy was one of the biggest challenges I faced and a deeper knowledge in that area would have helped me resolve the issue faster